this project. Note that the OLED can be controlled through two pins of any AVR MCU using software, leaving I2C pins for other applications or control.

An actual-size PCB layout for the handy Zener meter is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the PCB and connect 9V battery to it.

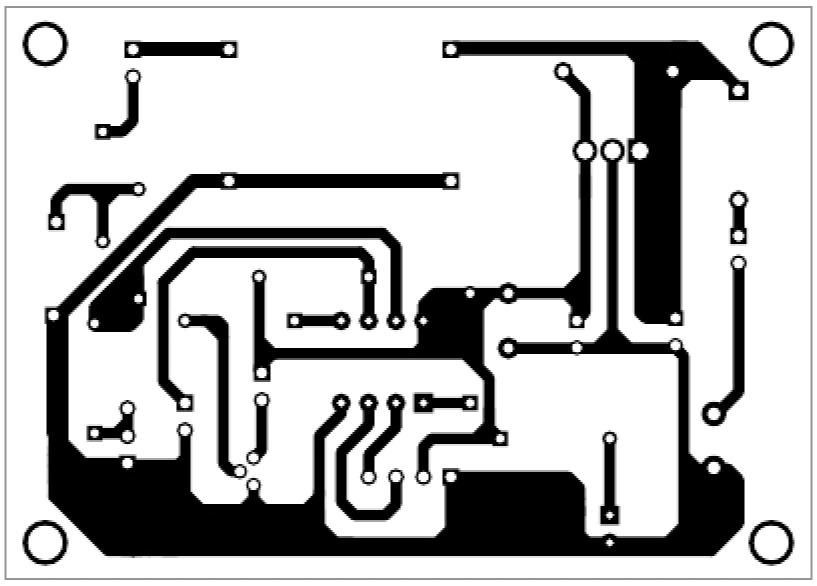

Fig. 2: Actual-size PCB layout of Zener meter

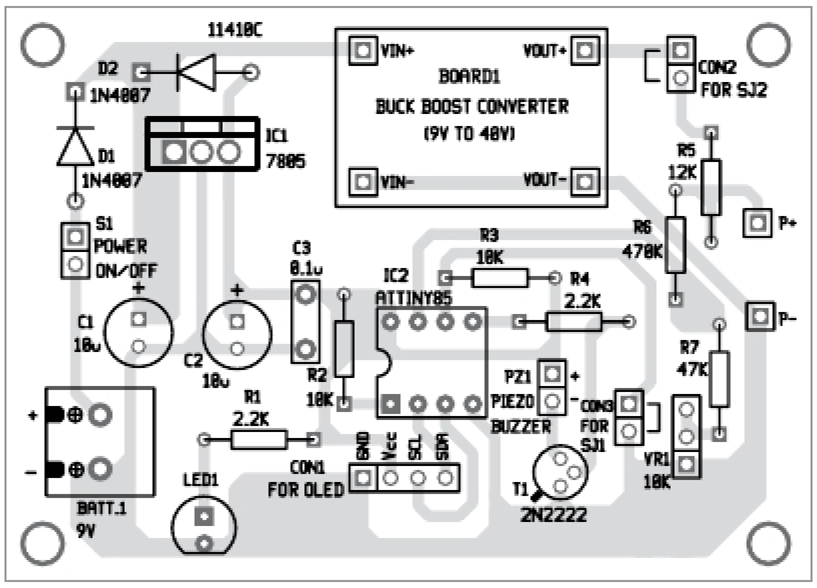

Fig. 3: Components layout of the PCB

Initial setup

1. Disconnect shorting jumpers SJ1 and SJ2. Write/program the zenermeter.hex code to ATtiny85 MCU through ISP port using an AVR programmer. Insert the programmed MCU on the PCB. It is recommended to use an 8-pin IC base for the MCU on the PCB.

2. Connect a multimeter in voltage mode across Vout+ and Vout- of the buck-boost module (Board1) and switch on S1. Then, adjust inbuilt potmeter of module to get about 42 volts. In case, the maximum output of the module is below 42 volts, then adjust the inbuilt potmeter to about 32 volts. Now, disconnect the multimeter.

3. Connect a known voltage source (between 10V DC and 25V DC, say 12V) and a multimeter (in voltage mode) across the probes P+ and P-. Adjust VR1 to match the voltage reading on OLED display with multimeter reading. Now, disconnect the multimeter.

4. Connect shorting jumpers SJ1 and SJ2. Now, restart the circuit by switching S1 off and on again. The 'Zener meter' along with a maximum voltage range message will appear on the OLED display. Connect the probes (P+ and P-) across a Zener diode, it will start displaying the unknown voltage drop of the diode. The circuit is now ready to use. This circuit can measure Zener voltages of up to 40V.

When no component is connected across the probes (P+ and P-), 'Zener absent' message is displayed on the OLED. When a Zener diode is connected (in reverse polarity), the voltage drop across the Zener diode is displayed on the OLED. When a junction diode is connected in forward bias mode, or both the probes are shorted (continuity tester mode), the voltage drop across the probes is displayed and a small beep is generated. In case an LED is connected in forward bias mode, the LED glows and voltage drop across the LED is also displayed. EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*

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